ANTLE RESIDENCE ADDITION

ELECTRICAL WIRE LAYOUT - DRAFT SET

3119 COLORADO AVE, ORLANDO, FL 32826 | ISSUED 2026-11-13 (DRAFT R1 - FIXTURES/DEVICES PER QUOTE)

PROJECT TEAM

PROJECT: ANTLE RESIDENCE ADDITION
SITE: 3119 COLORADO AVE, ORLANDO, FL 32826
OWNER: ANTLE (FULL OWNER NAME NOT ON PLANS - VERIFY)
GENERAL CONTRACTOR: NOK CONSTRUCTION
ELECTRICAL CONTRACTOR: SHARMA ELECTRIC & HVAC, EC13013924
ARCHITECTURAL BASE: ANTLE UPDATED PLANS DATED 10/29/2026 (BY OTHERS)

ELECTRICAL SCOPE (THIS SET)

- ONE-STORY ADDITION OF 314 SQ FT: NEW BEDROOM, BATHROOM, WALK-IN CLOSET, FOYER AND COVERED ENTRY.
- POWER COMES FROM THE EXISTING 150A SERVICE THROUGH THE EXISTING GE LOAD CENTER (TM1615CCU OR EQUAL - VERIFY). NO SUBPANEL AND NO SERVICE UPGRADE.
- TWO NEW CIRCUITS: A-1, 15A AREA CIRCUIT (14/2, 14/3 TO FAN, BATH AND SMOKE/CO) FOR ALL LIGHTS, FANS, GENERAL RECEPTACLES AND ALARMS; A-2, 20A BATH RECEPTACLE CIRCUIT (12/2) FOR THE VANITY AND BIDET RECEPTACLES ONLY.
- BREAKERS: GE THQL1115DF2 AND THQL1120DF2 DUAL-FUNCTION AFCI/GFCI (PER OWNER, VERIFY WITH SUPPLIER).
- R1 PER QUOTE: (7) AMICO 3" LED CANLESS (4 BEDROOM, 2 FOYER, 1 SHOWER - LISTING FOR SHOWER LOCATION TO BE VERIFIED), (3) LEVITON SURESLIDE DIMMERS ON BEDROOM AND BATH LIGHTS, (1) USB-A TR RECEPTACLE AT OWNER-SELECTED LOCATION, REGULAR TR DUPLEX AT VANITY.
- NO NEW AIR CONDITIONING, AIR HANDLER OR HEAT KIT. EXISTING HVAC STAYS AS IS.

CODE BASIS

FLORIDA BUILDING CODE 2023, 8TH EDITION (RESIDENTIAL)
NATIONAL ELECTRICAL CODE 2020 (NFPA 70) AS ADOPTED IN FLORIDA
SERVICE CHECK: NEC 220.83(A) EXISTING-DWELLING METHOD

GENERAL CONTRACTOR

NOK CONSTRUCTION

SHEET INDEX

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E-3.1	CIRCUIT A-1 - 15A AREA CIRCUIT	5 OF 6	R1 2026-11-13
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CLICK A ROW OR USE THE PDF BOOKMARKS TO JUMP TO A SHEET.

BUILDING DATA (FROM PLANS)

EXISTING TWO-STORY RESIDENCE: 1,752 SQ FT
NEW ONE-STORY ADDITION: 314 SQ FT
COMBINED LIVING AREA: 2,066 SQ FT (PLANS PRINT 2,067)
SERVICE: EXISTING 150A, 120/240V, 1-PHASE

SERVICE SUMMARY

NEC 220.83(A) CHECK: 21,359 VA = 89.0 A ON 150A - PASSES (SEE SHEET E-2).
EXISTING APPLIANCE NAMEPLATES ASSUMED ALL-ELECTRIC - VERIFY IN FIELD.
PANEL AND METER LOCATIONS ARE NOT ON THE PLANS - VERIFY IN FIELD.

SCHEMATIC KEY PLAN (NOT TO SCALE)

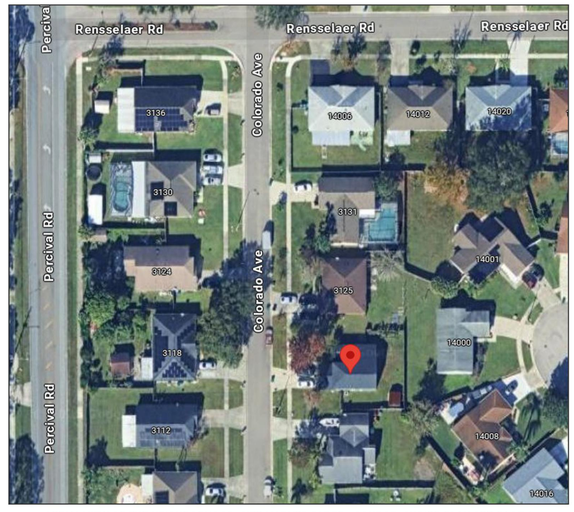


RELATIVE POSITIONS ONLY - SEE E-1.

PROJECT PHOTOS (FOR REFERENCE ONLY)



HOUSE RENDERING, FRONT VIEW - SOURCE: OWNER PLAN SET (10/29/2026), PAGE 1



AERIAL VIEW OF THE NEIGHBORHOOD, SITE MARKED IN RED - SOURCE: OWNER PLAN SET, PAGE 1

RED = POWER LINES. BLUE = SWITCH LEGS (14/3 AND 12/3 SWITCH LEGS = BLUE DOUBLE LINE). SMOKE/CO INTERCONNECT 14/3 = BLACK/YELLOW DASHED.

ELECTRICAL POWER DESIGN PREPARED BY:
SHARMA ELECTRIC & HVAC
6117 ARBOR WATCH LOOP, ORLANDO, FL 32829
407-881-9076 | OFFICE@SHARMAELECTRIC.COM
ELECTRICAL CONTRACTOR EC13013924
CERTIFIED AIR CONTRACTOR CAC1823596

PROJECT
ADDRESS
ANTLE RESIDENCE - ADDITION
3119 COLORADO AVE
ORLANDO, FL 32826
FEED: EXISTING GE TM1615CCU

CLIENT
NOK CONSTRUCTION

SHEET
TITLE
ELECTRICAL COVER SHEET & SHEET INDEX
SCALE
NTS
DATE
2026-11-13
REV R1 2026-11-13: FIXTURES/ DEVICES PER QUOTE
DRAWN BY
SHARMA ELECTRIC & HVAC

ELECTRICAL CONTRACTOR EC13013924 | CERTIFIED AIR CONTRACTOR CAC1823596

DRAFT
NOT FOR BUILDING DEPARTMENT RECORD

SHEET
CS
1 OF 6

SYMBOL / LINE LEGEND

- DEVICE (TAG PER CIRCUIT: R, SB, RL, F, VL, EF, SD/CO)
- POWER LINE (RED) - 14/2 ON 15A, 12/2 ON 20A
- HOME RUN TO EXISTING GE LOAD CENTER (RED, HEAVY, DIAGONAL IN ATTIC)
- SWITCH LEG (BLUE)
- 14/3 / 12/3 SWITCH LEG (BLUE DOUBLE LINE, LABELED)
- SMOKE/CO 14/3 INTERCONNECT (BLACK/YELLOW DASHED)
- EXISTING GE LOAD CENTER (VERIFY LOCATION IN FIELD)
- EXISTING METER (NO WORK, VERIFY LOCATION)

DEVICE / MATERIAL SCHEDULE (SHARMA STANDARD PARTS - RELEASE 3A)

TAG	DESCRIPTION	MANUFACTURER / CATALOG NO.	QTY
R1-R5, R7-R9, A2-R1	15A DUPLEX TR RECEPTACLE, WHITE (A2-R1 = REGULAR TR AT VANITY)	PASS & SEYMOUR 3232TRW / PLATE TP8W	9
R6	USB-A / DUPLEX TR RECEPT, 15A - LOCATION SELECTED BY OWNER	LEVITON T5632-W / PLATE TP26W	1
A2-R2	BIDET RECEPT., 20A DUPLEX TR (GFCI VIA BREAKER)	PASS & SEYMOUR 3232TR20SW - VERIFY LOC./HT. W/ BUILDER & BIDET SEAT	1
DIMMERS	SURESLIDE LED DIMMER 150W, 1P/3-WAY: SB2 SW-1, SB SW-1, SB SW-2	LEVITON 66741 (WHITE 66741-W)	3
ROCKERS	SINGLE-POLE DECORA ROCKER: SB1 x2, SB2 SW-2, SB3, SB SW-3	P&S TM870W	5
PLATES	SB3 1-G / SB1, SB2 2-G / SB 3-G DECORA	P&S TP26W / TP262W / TP263W	1 / 2 / 1
RL1-4, RL6, RL7-8	3 IN. LED CANLESS (4 BED, 1 SHOWER, 2 FOYER)	AMICO A18SP01AM1201001 - PER QUOTE, VERIFY	7
RL6 NOTE	SHOWER: LUMINAIRE LISTED FOR LOCATION, WET WHERE IN SPRAY (410.10(A).(D))	VERIFY AMICO LISTING BEFORE INSTALL	-
RL5 (CLOSET)	5 IN. CANLESS LED WAFER, IC (UNCHANGED)	AMICO A18SP01AM1004024 - VERIFY WITH BUILDER	1
EF	80 CFM BATH EXHAUST FAN	BROAN-NUTONE AE80B - VERIFY SUPPLIER	1
VL	VANITY LIGHT / VANITY BAR BOX	FIXTURE: VERIFY WITH BUILDER / BOX CARLON B620HG-UPC	1
F2	CEILING FAN (BY OWNER) / FAN-RATED BOX	FAN: VERIFY WITH BUILDER / BOX CENTAUR CM-FSLPB	1
F1	ENTRY CEILING FIXTURE / BOX	FIXTURE: VERIFY WITH BUILDER / BOX CES PSBR20	1
SD/CO1-2	HARDWIRED SMOKE/CO COMBO, BATT. BACKUP, INTERCONN.	BRK SMICO100-AC	2
WALL BOXES	1-G 22 / 2-G 32 / 3-G 44 CU IN NEW-WORK	CES PSB122 / PSB232 / PSB344	12 / 2 / 1
CEILING BOXES	4 IN. ROUND 20 CU IN	CES PSBR20	AS REQ.
BREAKERS	DUAL-FUNCTION AFCI/GFCI, A-1 15A 1P / A-2 20A 1P	GE THQL1115DF2 / GE THQL1120DF2 - PER OWNER, VERIFY WITH SUPPLIER	1 / 1
CABLE	14/2, 14/3 (A-1: FAN, BATH, SMOKE/CO); 12/2 (A-2) NM-B CU	CES NM-B-14/2-CU-250C, NM-B-14/3-CU-250C, NM-B-12/2-CU-250C	SEE E-2
NAIL PLATES	16 GA STEEL KICK PLATES (300.4)	MINERALLAC 90100	31
STAPLES	NM INSULATED STAPLES	F4P S100-100	1 JAR

GENERAL NOTES

1. SCOPE: NEW ADDITION ONLY (NEW BEDROOM, BATHROOM, WALK-IN CLOSET, FOYER AND COVERED ENTRY, 314 SQ FT). EXISTING HOUSE WIRING NOT IN SCOPE EXCEPT CONNECTION AT THE EXISTING PANEL.
2. CODE BASIS: NEC 2020 AS ADOPTED BY THE FLORIDA BUILDING CODE 8TH EDITION (2023). ARCHITECT SHEET NOTES CITE NEC 2008; NEC 2020 GOVERNS THIS DESIGN.
3. DEVICE LOCATIONS FOLLOW ARCHITECT SHEET 04 "PROPOSED ELEC. LAYOUT PLAN". ITEMS ADDED BY SHARMA STANDARD ARE CALLED OUT ON E-1.
4. FEED: ALL NEW CIRCUITS HOME-RUN TO THE EXISTING GE LOAD CENTER ON THE EXISTING 150A SERVICE. NO SUBPANEL. GE TM1615CCU (150A MCB, 16 SP/32 CKT) OR RELIAHOME PAM1615 EQUIVALENT. VERIFY MODEL, LOCATION & OPEN SPACES IN FIELD.
5. ALL NEW 120V CIRCUITS USE DUAL-FUNCTION AFCI/GFCI BREAKERS MATCHING THE EXISTING GE PANEL: A-1 GE THQL1115DF2, A-2 GE THQL1120DF2. NO GFI RECEPTACLES.
6. ALL RECEPTACLES TAMPER-RESISTANT (406.12). RECEPTACLES 18" AFF TO BOTTOM OF BOX UNLESS THE PLAN STATES A HEIGHT (PLAN GOVERNS).
7. NM-B COPPER. 14 AWG ON 15A (240.4(D)), 12 AWG ON 20A. ARCHITECT NOTE 13 CALLS #12 MIN.; SHARMA USING 14/2 ON 15A PER NEC 240.4(D) - BUILDER TO CONFIRM ACCEPTABLE.
8. KICK PLATES: 16 GA STEEL NAIL PLATES AT EVERY TOP-PLATE DROP (300.4); CENTER-BORED STUDS NEED NO PLATES. COUNT = DROPS + 10%.
9. NO WIRING WITHIN 6 FT OF THE ATTIC SCUTTLE; NM WITHIN 6 FT OF THE SCUTTLE GETS GUARD STRIPS OR BORED HOLES.
10. SMOKE/CO ALARMS: HARDWIRED, BATTERY BACKUP, INTERCONNECTED. ONE FEED INTO ONE ALARM; OTHERS DAISY-CHAIN ON 14/3.
11. NO NEW A/C, AIR HANDLER OR HEAT KIT CIRCUITS IN THIS SCOPE.
12. DRAFT FOR REVIEW. NOT FOR BUILDING DEPARTMENT RECORD.

SHEET INDEX

CS ELECTRICAL COVER SHEET & SHEET INDEX
E-0 LEGEND, GENERAL NOTES & SHEET INDEX
E-1 BRANCH CIRCUIT PLAN - ALL NEW CIRCUITS (ADDITION)
E-2 NEW CIRCUIT SUMMARY & NEC 220 LOAD CALC (ADDITION)
E-3.1 CIRCUIT A-1 - 15A AREA CIRCUIT
E-3.2 CIRCUIT A-2 - 20A BATH RECEPTACLES

DRAFT

RED = POWER LINES. BLUE = SWITCH LEGS (14/3 AND 12/3 SWITCH LEGS = BLUE DOUBLE LINE). SMOKE/CO INTERCONNECT 14/3 = BLACK/YELLOW DASHED.

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PROJECT

ADDRESS

ANTLE RESIDENCE - ADDITION
3119 COLORADO AVE
ORLANDO, FL 32826
FEED: EXISTING GE TM1615CCU

CLIENT

NOK CONSTRUCTION

SHEET

TITLE

LEGEND, GENERAL NOTES & SHEET INDEX

SCALE
NTS

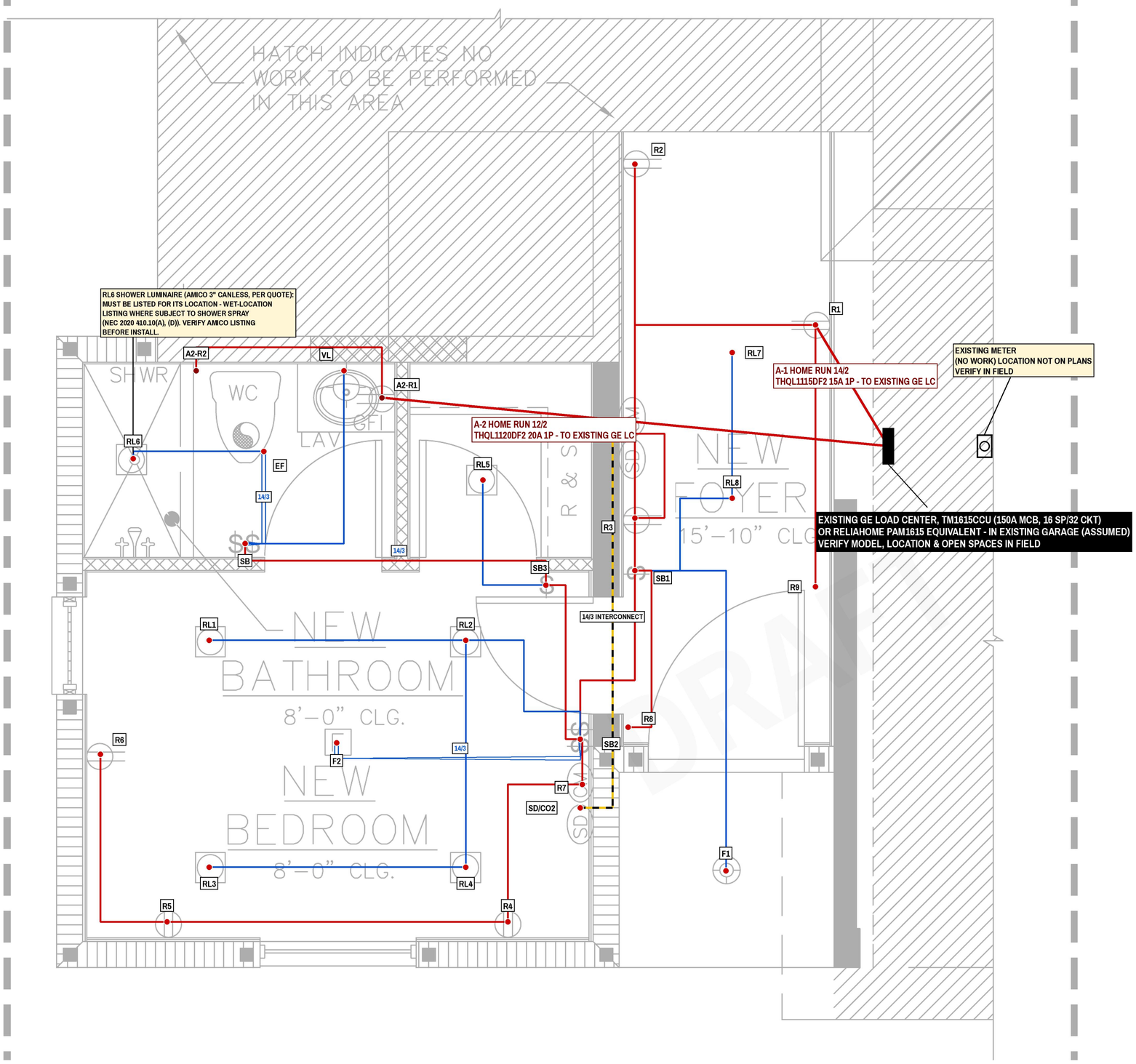
DATE
2026-11-13
REV R1 2026-11-13: FIXTURES/ DEVICES PER QUOTE

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SHEET
E-0
2 OF 6



CIRCUIT LEGEND (NEW CIRCUITS IN EXISTING GE LOAD CENTER)

- **A-1 15A 1P AREA CIRCUIT - BEDROOM, CLOSET, FOYER, ENTRY, ALL BATH LIGHTS + EXHAUST FAN, SMOKE/CO**
  GE THQL1115DF2 15A 1P DUAL-FUNCTION AFCI/GFCI | 14/2 NM-B CU (14/3 TO FAN, BATH, SMOKE/CO) - SEE NOTE 14
- **A-2 20A 1P BATHROOM RECEPTACLES ONLY - VANITY GFCI RECEPT. + BIDET RECEPT.**
  GE THQL1120DF2 20A 1P DUAL-FUNCTION AFCI/GFCI | 12/2 NM-B CU
- **SWITCH LEG (BLUE)**
- **14/3 OR 12/3 SWITCH LEG (BLUE DOUBLE LINE)**
- **SMOKE/CO 14/3 INTERCONNECT (BLACK/YELLOW DASHED)**

KEYED NOTES

1. ADDITION FEEDS FROM THE EXISTING GE LOAD CENTER ON THE EXISTING 150A SERVICE (NO SUBPANEL). EXISTING GE LOAD CENTER, TM1615CCU (150A MCB, 16 SP/32 CKT) OR RELIAHOME PAM1615 EQUIVALENT. LOCATION IS NOT ON THE PLANS; SHOWN IN THE EXISTING GARAGE AS ASSUMED. VERIFY MODEL, LOCATION & OPEN SPACES IN FIELD.
2. CIRCUIT NUMBERS A-1 / A-2 ARE ADDITION IDs. ASSIGN THE ACTUAL GE LOAD CENTER SPACE NUMBERS IN FIELD AND UPDATE THE PANEL DIRECTORY.
3. NEW BREAKERS MUST MATCH THE EXISTING GE PANEL: A-1 GE THQL1115DF2 (15A 1P) AND A-2 GE THQL1120DF2 (20A 1P), DUAL-FUNCTION AFCI/GFCI. NO GFI RECEPTACLES (DUAL-FUNCTION BREAKERS).
4. HOME RUNS ARE DRAWN DIAGONAL THROUGH THE ATTIC; ALL OTHER ATTIC RUNS PARALLEL/PERPENDICULAR TO TRUSSES. EXTERIOR WALLS ARE NEW CMU WITH P.T. FURRING: DROPS COME DOWN FROM THE ATTIC IN THE FURRING SPACE (NO BOX-TO-BOX BORING IN CMU).
5. FOYER CEILING IS 15'-10". CONFIRM ATTIC ACCESS ABOVE THE FOYER FOR DROPS (DRAWN AS ATTIC RUNS).
6. NO ATTIC SCUTTLE IS SHOWN IN THE ADDITION. NO WIRING WITHIN 6 FT OF ANY ATTIC SCUTTLE; NM WITHIN 6 FT OF THE SCUTTLE GETS GUARD STRIPS OR BORED HOLES (FIELD VERIFY EXISTING SCUTTLE LOCATION).
7. SMOKE/CO: ONE 120V FEED INTO SD/CO1 (FOYER); SD/CO2 (BEDROOM) DAISY-CHAINED ON 14/3. PLAN SHOWS SEPARATE SD + CM; SINGLE COMBO UNITS USED PER ARCHITECT NOTE 9. EXISTING-HOUSE ALARM INTERCONNECT: VERIFY IN FIELD.
8. A-2 (20A, 12/2, THQL1120DF2) SERVES ONLY THE BATH RECEPTACLES: A2-R1 AT THE VANITY AND A2-R2 BIDET RECEPTACLE BESIDE THE WC (~6-8 IN TO SIDE, ~12-18 IN AFF - VERIFY LOCATION/HEIGHT WITH BUILDER & BIDET SEAT). GFCI PROTECTION FROM THE DUAL-FUNCTION BREAKER (210.8(A)(1), 210.11(C)(3)). ALL BATH LIGHTS AND THE EXHAUST FAN ARE ON 15A A-1, FED 14/3 FROM SB3 IN THE BEDROOM TO THE BATH 3-GANG SB (A-1 HOME RUN DOES NOT LAND IN THE BATH). SB (3-GANG): SW-1 = VL (SURESLIDE DIMMER, 14/2 LEG); SW-2 = SHOWER RL6 (SURESLIDE DIMMER); SW-3 = EF (ROCKER). 14/3 FROM SB TO EF CARRIES THE EF AND RL6 SWITCHED LEGS; 14/2 EF TO RL6.
9. BATH EXHAUST FAN (EF), VANITY LIGHT (VL) AND BIDET RECEPTACLE ARE NOT DRAWN BY THE ARCHITECT; EF/VL ADDED PER SHARMA STANDARD, BIDET PER OWNER - VERIFY WITH BUILDER. ARCHITECT SHOWS 2 BATH SWITCHES; R1 USES A 3-GANG SO THE SHOWER LIGHT CAN BE DIMMED SEPARATELY FROM THE FAN (NO DIMMER ON A FAN MOTOR).
10. ARCHITECT SHOWS NO SWITCH ARCS; SWITCH LEGS ARE DRAWN TO THE NEAREST ARCHITECT SWITCH SYMBOL - VERIFY. SB1 (FOYER, 2-GANG): SW-1 ENTRY FIXTURE F1, SW-2 FOYER CANS RL7/RL8 (R1, PER QUOTE; NOT ON ARCHITECT PLAN). ROCKERS.
11. NO SWITCHED (HALF-HOT) RECEPTACLES. RECEPTACLES 18" AFF TO BOTTOM OF BOX UNLESS PLAN STATES OTHERWISE; VANITY RECEPTACLE (A2-R1) ABOVE VANITY WITHIN 3 FT OF BASIN - HEIGHT VERIFY.
12. NO NEW A/C, AIR HANDLER OR HEAT KIT IN THIS SCOPE (EXISTING HVAC TO REMAIN).
13. RECEPTACLES R7-R9 ADDED PER NEC 210.52(A)(1)/(A)(2) (NO POINT ALONG WALL > 6 FT FROM A RECEPT.): R7 BEDROOM WALL BESIDE SB2 AT THE BEDROOM DOOR; R8 FOYER 2+ FT WALL SECTION BESIDE THE ENTRY DOOR; R9 FOYER EAST WALL (AT EDGE OF ENTRY DOOR SWING - VERIFY). WALK-IN CLOSET NOT A 210.52(A) ROOM - NONE ADDED. A-1 NOW ~2,010 VA (OVER ~1,800 VA TARGET) - NOT SPLIT, FOR SANJAY TO DECIDE.
14. R1: FIXTURES/DEVICES PER QUOTE - (7) AMICO 3" LED CANLESS A18SP01AM1201001 (PER QUOTE, VERIFY): RL1-RL4 BEDROOM, RL7-RL8 FOYER, RL6 SHOWER (REPLACES VAPOR-PROOF FIXTURE; SEE SHOWER NOTE AT RL6 - LISTING MUST SUIT THE LOCATION, WET WHERE SUBJECT TO SHOWER SPRAY, NEC 2020 410.10(A), (D); VERIFY AMICO LISTING BEFORE INSTALL). CLOSET RL5 UNCHANGED. (3) LEVITON SURESLIDE 06741 DIMMERS: BEDROOM CANS, BATH VL, SHOWER RL6; ALL OTHER SWITCHES ROCKERS. USB-A TR RECEPT. (LEVITON T5632-W) AT R6, LOCATION SELECTED BY OWNER. VANITY A2-R1 = REGULAR TR DUPLEX (GFCI VIA BREAKER).
15. ARCHITECT NOTE 13 CALLS #12 MIN.; SHARMA USING 14/2 ON THE 15A CIRCUIT A-1 (14/3 TO CEILING FAN, BATH SB/EF AND SMOKE/CO INTERCONNECT) PER NEC 240.4(D). A-2 (20A BATH RECEPTACLES) IS 12/2. BUILDER TO CONFIRM ACCEPTABLE.

1 BRANCH CIRCUIT PLAN - ALL NEW CIRCUITS

SCALE: NTS (BASE = ARCHITECT SHEET 04 PROPOSED ELEC. LAYOUT PLAN) | NEW ADDITION ONLY - EXISTING HOUSE NOT IN SCOPE

RED = POWER LINES. BLUE = SWITCH LEGS (14/3 AND 12/3 SWITCH LEGS = BLUE DOUBLE LINE). SMOKE/CO INTERCONNECT 14/3 = BLACK/YELLOW DASHED.

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PROJECT

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FEED: EXISTING GE TM1615CCU

CLIENT

NOK CONSTRUCTION

SHEET

TITLE
BRANCH CIRCUIT PLAN - ALL NEW CIRCUITS (ADDITION)

SCALE
NTS (BASE = ARCHITECT SHEET 04)

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2026-11-13
REV R1 2026-11-13: FIXTURES/ DEVICES PER QUOTE

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NOT FOR BUILDING DEPARTMENT RECORD

SHEET
E-1
3 OF 6

NEW CIRCUITS - EXISTING GE LOAD CENTER (NO PANEL SCHEDULE IN THIS SET)

CKT	DESCRIPTION	BREAKER	TYPE	WIRE	CONNECTED
A-1	AREA: 7 AMICO CANS (4 BED, 2 FOYER, 1 SHOWER), FAN, CLOSET RL, 9 RECEPT. (1 USB), ENTRY LT, BATH VL/EF, 2 SMOKE/CO	THQL1115DF2 15A 1P	DF AFCI/GFCI	14/2	~2,010 VA *
A-2	BATH RECEPTACLES ONLY: VANITY REGULAR TR DUPLEX + BIDET RECEPT. (GFCI VIA BREAKER)	THQL1120DF2 20A 1P	DF AFCI/GFCI	12/2	~360 VA

EXISTING GE TM1615CCU (150A MCB, 16 SP/32 CKT) OR RELIAHOME PAM1615 EQUIV. - VERIFY MODEL, LOCATION & OPEN SPACES IN FIELD. SPACE NUMBERS: ASSIGN IN FIELD. 2 NEW 1P SPACES. A-1 14/2 PER 240.4(D) (ARCH. NOTE 13 #12 - CONFIRM).

*** A-1 CONNECTED ~2,010 VA (9 RECEPT. x 180 = 1,620 + LIGHTS/FANS/ALARMS ~390) EXCEEDS THE ~1,800 VA SHARMA AREA-CIRCUIT TARGET - FLAGGED FOR SANJAY, CIRCUIT NOT SPLIT.**

CABLE FOOTAGE - ALL NEW CIRCUITS (FT)

CABLE	BEFORE WASTE	AFTER +10% WASTE
12/2	53	58
14/2	441	485
14/3	68	75

ADDITION LOAD BASIS (NEC 220)

1. NEW LIVING AREA (ARCHITECT SQ FT CALCS): 314 SQ FT. EXISTING LIVING SPACE 1,752 SQ FT; TOTAL 2,067 SQ FT (PER PLAN; NOTE 1,752 + 314 = 2,066).
2. GENERAL LIGHTING & RECEPTACLE LOAD OF ADDITION, NEC 220.12 / TABLE 220.12: 314 SQ FT x 3 VA = 942 VA.
3. NO NEW SMALL-APPLIANCE, LAUNDRY, FIXED APPLIANCE, HEATING OR A/C LOAD IN THE ADDITION.
4. EXISTING SERVICE: 150A, 120/240V (PER OWNER). EXISTING-DWELLING CALCULATION PER NEC 220.83(A) BELOW (NO NEW A/C OR ELECTRIC HEAT).

NEC 2020 220.83(A) LOAD CALC - EXISTING 150A SERVICE + ADDITION (NO NEW A/C OR ELECTRIC HEAT)

LOAD	VA	BASIS
GENERAL LIGHTING & RECEPT. - 2,066 SQ FT x 3 VA (1,752 EXIST. + 314 ADDITION)	6,198	PLANS SHT 02
SMALL-APPLIANCE CIRCUITS - 2 x 1,500 VA (EXISTING KITCHEN)	3,000	220.83
LAUNDRY CIRCUIT - 1,500 VA	1,500	ASSUMED - VERIFY
ELECTRIC RANGE (NAMEPLATE)	12,000	ASSUMED - VERIFY
ELECTRIC DRYER (NAMEPLATE)	5,000	ASSUMED - VERIFY
ELECTRIC WATER HEATER (NAMEPLATE)	4,500	ASSUMED - VERIFY
DISHWASHER (NAMEPLATE)	1,200	ASSUMED - VERIFY
EXISTING A/C, CONDENSER + AIR HANDLER (ALLOWANCE, NOT NAMEPLATE)	8,000	ASSUMED - VERIFY
CONNECTED LOAD	41,398	
FIRST 8,000 VA AT 100%	8,000	220.83(A)
REMAINDER 33,398 VA AT 40%	13,359	220.83(A)
CALCULATED DEMAND	21,359 VA	
CALCULATED LOAD = 21,359 / 240V	89.0 A	

RESULT: 89.0 A vs EXISTING 150A SERVICE = PASS (59% OF RATING, 61 A HEADROOM)

1. EXISTING APPLIANCE NAMEPLATES ASSUMED ALL-ELECTRIC (RANGE, DRYER, WATER HEATER, DISHWASHER, LAUNDRY, A/C ALLOWANCE) - NONE ARE ON THE PLANS. VERIFY IN FIELD.
2. HEAT-KIT CHECK: CONSERVATIVE CASE + 10 KW ELECTRIC HEAT KIT = 51,398 VA CONNECTED, 25,359 VA DEMAND = 105.7 A - ALSO PASSES ON 150A.
3. ADDITION ADDS 314 SQ FT x 3 VA = 942 VA GENERAL LIGHTING ONLY; NO NEW 240V, APPLIANCE OR HVAC LOAD. CIRCUITS A-1 AND A-2 ARE INCLUDED IN THE 3 VA/SQ FT LOAD.
4. PLAN PRINTS 2,067 SQ FT TOTAL; 1,752 + 314 = 2,066 SQ FT USED (3 VA DIFFERENCE, NO EFFECT).

DRAFT

RED = POWER LINES. BLUE = SWITCH LEGS (14/3 AND 12/3 SWITCH LEGS = BLUE DOUBLE LINE). SMOKE/CO INTERCONNECT 14/3 = BLACK/YELLOW DASHED.

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FEED: EXISTING GE TM1615CCU

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NOK CONSTRUCTION

SHEET

TITLE
NEW CIRCUIT SUMMARY &
NEC 220 LOAD CALC
(ADDITION)

SCALE
NTS

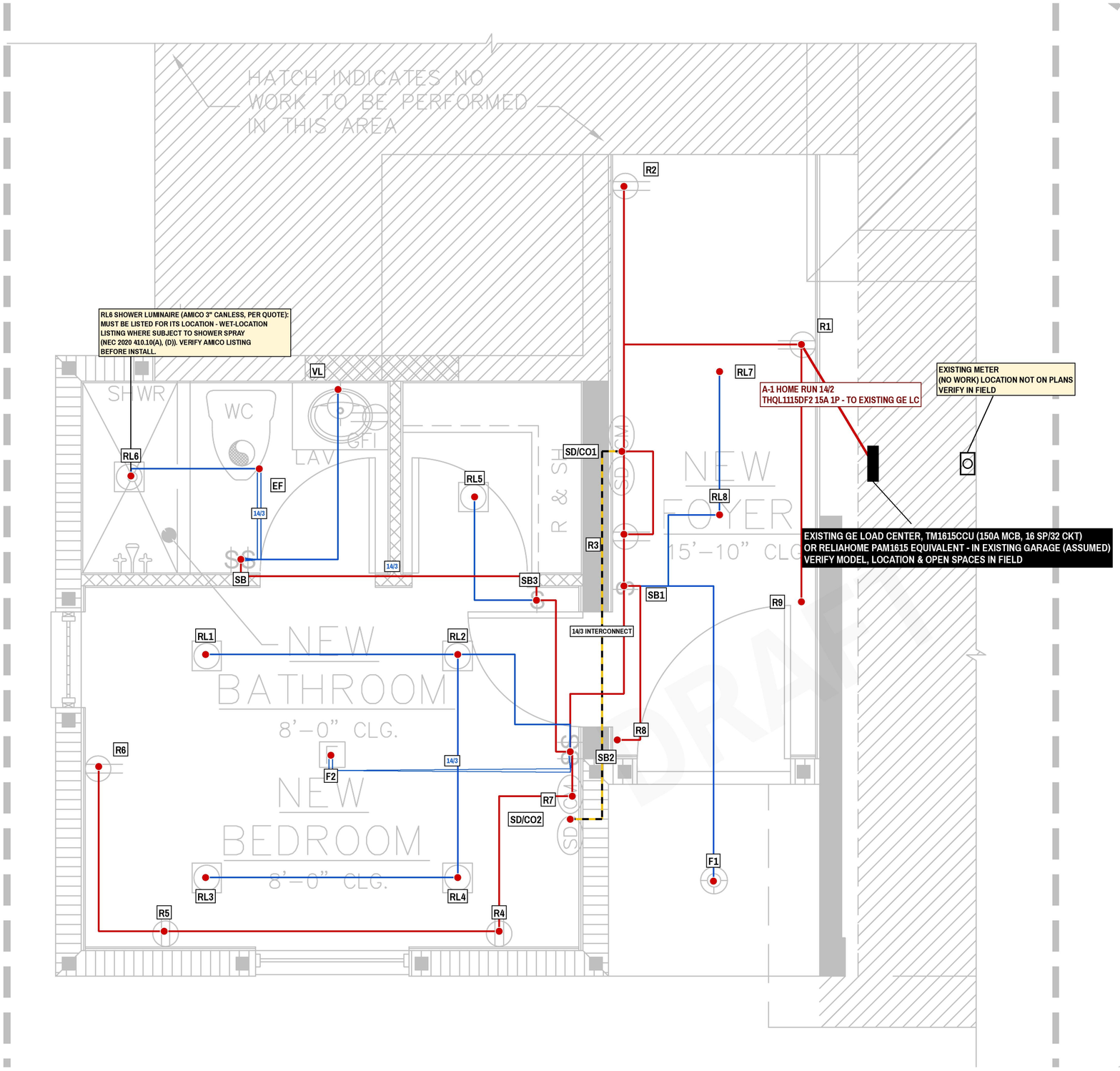
DATE
2026-11-13
REV R1 2026-11-13: FIXTURES/
DEVICES PER QUOTE

DRAWN BY
SHARMA ELECTRIC & HVAC

ELECTRICAL CONTRACTOR EC13013924 | CERTIFIED AIR CONTRACTOR CAC1823596

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SHEET
E-2
4 OF 6



RUN SCHEDULE - A-1

#	FROM	TO	TYPE	ROUTE	CABLE	FT
1	PANEL	R1	POWER	ATTIC DROP	14/2	25.5
2	R1	R2	POWER	ATTIC DROP	14/2	43.6
3	R2	R3	POWER	BORED STUDS	14/2	15.4
4	R3	SB1	POWER	BORED STUDS	14/2	4.0
5	SB1	F1	SWITCH LEG	ATTIC DROP	14/2	28.6
6	R3	SD/CO1	POWER	ATTIC DROP	14/2	21.8
7	SD/CO1	SD/CO2	SMOKE INTERCONNECT	ATTIC DROP	14/3	18.1
8	SB1	SB2	POWER	ATTIC DROP	14/2	26.3
9	SB1	RL8	SWITCH LEG	ATTIC DROP	14/2	20.2
10	RL8	RL7	SWITCH LEG	ATTIC DROP	14/2	7.5
11	SB2	SB3	POWER	BORED STUDS	14/2	9.1
12	SB3	RL5	SWITCH LEG	ATTIC DROP	14/2	12.4
13	SB2	RL2	SWITCH LEG	ATTIC DROP	14/2	14.1
14	RL2	RL1	SWITCH LEG	ATTIC DROP	14/2	11.7
15	RL2	RL4	SWITCH LEG	ATTIC DROP	14/2	10.6
16	RL4	RL3	SWITCH LEG	ATTIC DROP	14/2	11.7
17	SB2	F2	SWITCH LEG	ATTIC DROP	14/3	16.5
18	SB2	R7	POWER	BORED STUDS	14/2	3.8
19	R7	R4	POWER	ATTIC DROP	14/2	23.0
20	SB1	R8	POWER	ATTIC DROP	14/2	35.6
21	R1	R9	POWER	ATTIC DROP	14/2	40.6
22	R4	R5	POWER	ATTIC DROP	14/2	27.9
23	R5	R6	POWER	ATTIC DROP	14/2	23.9
24	SB3	SB	POWER	ATTIC DROP	14/3	23.0
25	SB	VL	SWITCH LEG	ATTIC DROP	14/2	16.3
26	SB	EF	SWITCH LEG	ATTIC DROP	14/3	10.2
27	EF	RL6	SWITCH LEG	ATTIC DROP	14/2	7.2

FOOTAGE (FT, DROPS INCL., +10% WASTE ONCE)

CABLE	BEFORE WASTE	+10%	AFTER WASTE
14/2	441	44	485
14/3	68	7	75

DEVICES ON THIS CIRCUIT

TAG	DEVICE
R1	DUPLEX TR RECEPT. 18" AFF
R2	DUPLEX TR RECEPT. 18" AFF
R3	DUPLEX TR RECEPT. 18" AFF
SB1	2-GANG SB (FOYER): SW-1 F1 ROCKER, SW-2 RL7/RL8 ROCKER
F1	ENTRY CEILING FIXTURE
SD/CO1	SMOKE/CO COMBO
SD/CO2	SMOKE/CO COMBO
SB2	2-GANG SB (BEDROOM): SW-1 CANS SURESLIDE DIMMER, SW-2 FAN ROCKER
SB3	1-GANG SWITCH BOX (CLOSET)
RL1	AMICO 3" LED CANLESS (PER QUOTE)
RL2	AMICO 3" LED CANLESS (PER QUOTE)
RL3	AMICO 3" LED CANLESS (PER QUOTE)
RL4	AMICO 3" LED CANLESS (PER QUOTE)
RL5	RECESSED LIGHT (CLOSET)
F2	CEILING FAN (FAN-RATED BOX)
RL7	AMICO 3" LED CANLESS (FOYER, PER QUOTE)
RL8	AMICO 3" LED CANLESS (FOYER, PER QUOTE)
R4	DUPLEX TR RECEPT. 18" AFF
R5	DUPLEX TR RECEPT. 18" AFF
R6	USB-A/DUPLEX TR RECEPT. (LEVITON T5632-W) - LOCATION SELECTED BY OWNER
R7	DUPLEX TR RECEPT. 18" AFF (BESIDE SB2, 210.52(A))
R8	DUPLEX TR RECEPT. 18" AFF (BESIDE ENTRY DOOR, 210.52(A))
R9	DUPLEX TR RECEPT. 18" AFF (FOYER E. WALL, 210.52(A))
SB	3-GANG SB (BATH): SW-1 VL DIMMER, SW-2 RL6 DIMMER, SW-3 EF ROCKER
VL	VANITY LIGHT OVER MIRROR
RL6	AMICO 3" LED CANLESS (SHOWER, PER QUOTE) - SEE SHOWER NOTE
EF	BATH EXHAUST FAN

SHEET NOTES

1. HOME RUN TO THE EXISTING GE LOAD CENTER (LOCATION ASSUMED - VERIFY IN FIELD). BREAKERS: A-1 GE THQL1115DF2, A-2 GE THQL1120DF2.
2. ONLY THIS CIRCUIT IS SHOWN. SEE E-0 FOR LEGEND AND GENERAL NOTES, E-1 FOR ALL CIRCUITS.
3. FOOTAGE = PLAN LENGTH AT SHEET 04 SCALE + DROPS (8'-0" CLG BEDROOM/BATH, 15'-10" FOYER, 8'-4" GARAGE) + 1 FT MAKEUP PER END. FIELD VERIFY.

1 CIRCUIT A-1 - 15A AREA CIRCUIT

GE THQL1115DF2 15A 1P DUAL-FUNCTION AFCI/GFCI | 14/2 NM-B (14/3 TO FAN, BATH + SMOKE/CO) | EXISTING GE TM1615CCU, SPACE: ASSIGN IN FIELD

RED = POWER LINES. BLUE = SWITCH LEGS (14/3 AND 12/3 SWITCH LEGS = BLUE DOUBLE LINE). SMOKE/CO INTERCONNECT 14/3 = BLACK/YELLOW DASHED.

ELECTRICAL POWER DESIGN PREPARED BY:

SHARMA ELECTRIC & HVAC

6117 ARBOR WATCH LOOP, ORLANDO, FL 32829
407-881-9076 | OFFICE@SHARMAELECTRIC.COM
ELECTRICAL CONTRACTOR EC13013924
CERTIFIED AIR CONTRACTOR CAC1823596

PROJECT

ADDRESS

ANTLE RESIDENCE - ADDITION
3119 COLORADO AVE
ORLANDO, FL 32826
FEED: EXISTING GE TM1615CCU

CLIENT

NOK CONSTRUCTION

SHEET

TITLE
CIRCUIT A-1 - 15A AREA CIRCUIT

SCALE
NTS (BASE = ARCHITECT SHEET 04)

DATE
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REV R1 2026-11-13: FIXTURES/ DEVICES PER QUOTE

DRAWN BY
SHARMA ELECTRIC & HVAC

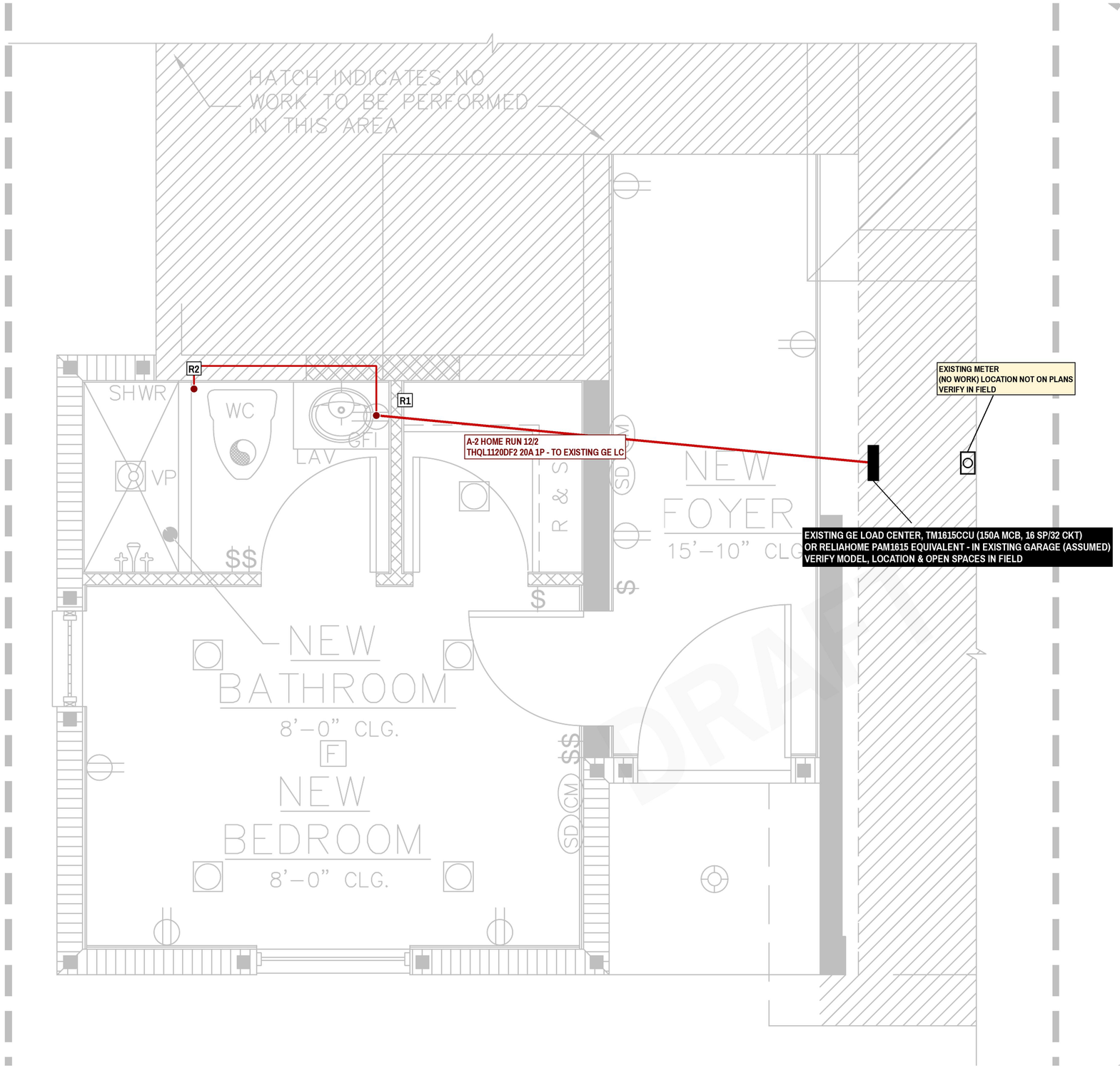
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NOT FOR BUILDING DEPARTMENT RECORD

SHEET
E-3.1

5 OF 6



RUN SCHEDULE - A-2

#	FROM	TO	TYPE	ROUTE	CABLE	FT
1	PANEL	R1	POWER	ATTIC DROP	12/2	29.6
2	R1	R2	POWER	ATTIC DROP	12/2	23.1

FOOTAGE (FT, DROPS INCL., +10% WASTE ONCE)

CABLE	BEFORE WASTE	+10%	AFTER WASTE
12/2	53	5	58

DEVICES ON THIS CIRCUIT

TAG	DEVICE
R1	REGULAR TR DUPLEX AT VANITY (GFCI VIA THQL1120DF2)
R2	BIDET RECEPT. 12-18" AFF, 6-8" SIDE OF WC - VERIFY

SHEET NOTES

1. HOME RUN TO THE EXISTING GE LOAD CENTER (LOCATION ASSUMED - VERIFY IN FIELD). BREAKERS: A-1 GE THQL1115DF2, A-2 GE THQL1120DF2.
2. ONLY THIS CIRCUIT IS SHOWN. SEE E-0 FOR LEGEND AND GENERAL NOTES, E-1 FOR ALL CIRCUITS.
3. FOOTAGE = PLAN LENGTH AT SHEET 04 SCALE + DROPS (8'-0" CLG BEDROOM/BATH, 15'-10" FOYER, 8'-4" GARAGE) + 1 FT MAKEUP PER END. FIELD VERIFY.

1 CIRCUIT A-2 - 20A BATH RECEPTACLES

GE THQL1120DF2 20A 1P DUAL-FUNCTION AFCI/GFCI | 12/2 NM-B | EXISTING GE TM1615CCU, SPACE: ASSIGN IN FIELD

RED = POWER LINES. BLUE = SWITCH LEGS (14/3 AND 12/3 SWITCH LEGS = BLUE DOUBLE LINE). SMOKE/CO INTERCONNECT 14/3 = BLACK/YELLOW DASHED.

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PROJECT

ADDRESS
ANTLE RESIDENCE - ADDITION
3119 COLORADO AVE
ORLANDO, FL 32826
FEED: EXISTING GE TM1615CCU

CLIENT

NOK CONSTRUCTION

SHEET

TITLE
CIRCUIT A-2 - 20A BATH RECEPTACLES

SCALE
NTS (BASE = ARCHITECT SHEET 04)

DATE
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NOT FOR BUILDING DEPARTMENT RECORD

SHEET
E-3.2